Embedded C Simulation Report

Title

Smart Home Lighting System Using Motion and Ambient Sensing

Abstract

This project presents a smart home lighting system implemented as an Embedded C simulation. The objective is to improve energy efficiency by automatically controlling room lighting based on human presence and surrounding light intensity. A PIR motion sensor is used to detect occupancy, while an LDR sensor measures ambient light. The controller turns the LED ON only when motion is detected in a dark environment. This simple automation concept is widely used in modern smart homes, offices, hospitals and public buildings.

Objectives

• Automate lighting control.

• Reduce energy consumption.

• Improve user convenience.

• Demonstrate embedded programming concepts.

• Simulate sensor-based decision making.

Requirements Analysis

Inputs: PIR motion sensor and LDR sensor.

Output: LED.

Functional logic: LED turns ON only if motion is detected and the room is dark (LDR threshold below 500). Otherwise, the LED remains OFF.

System Design

The simulation continuously reads two inputs. The Embedded C program compares the sensor values and makes a decision. This mimics the operation of a microcontroller in a real embedded system.

Algorithm

1. Start.
2. Initialize variables.
3. Read PIR value.
4. Read LDR value.
5. If motion is detected and LDR<500, turn LED ON.
6. Else turn LED OFF.
7. Repeat continuously.

Implementation

The program is written in Embedded C and can be compiled using Code::Blocks or Keil uVision. It uses standard input/output to simulate the sensor readings.

Testing and Validation

Four test cases were performed to verify the functionality. The LED turned ON only when both conditions were satisfied. All test cases passed successfully, validating the logic of the simulation.

Result Analysis

The system behaves correctly under different lighting and motion conditions. The design minimizes unnecessary energy consumption while maintaining user comfort.

Applications

Smart homes, offices, classrooms, hospitals, parking areas, hotels, warehouses and energy-efficient buildings.

Advantages

Automatic operation, reduced electricity bills, easy implementation, scalable design, improved energy efficiency and user convenience.

Limitations

The console simulation does not interface with physical hardware. Sensor values are manually entered by the user.

Future Enhancements

Integrate ESP32/Arduino hardware, IoT monitoring, mobile application control, voice assistants, PWM brightness control and cloud analytics.

Conclusion

The Embedded C simulation successfully demonstrates a smart home lighting system using motion and ambient sensing. The project is simple, reliable and suitable for understanding embedded system design principles.

CODE:

```
C main.c 2 ×
C: > Users > kandi > Downloads > Embedded systems Assessments > Simulation > C main.c > main()

int main()
{
    int motion;
    int ldr;

    printf("SMART HOME LIGHTING SYSTEM\n");

    while(1)
    {
        printf("\nEnter Motion (1=Detected,0=No Motion): ");
        scanf("%d",&motion);

        printf("Enter LDR Value (0-1023): ");
        scanf("%d",&ldr);

        if(motion==1 && ldr<500)
        {
            printf("Room is Dark\n");
            printf("Motion Detected\n");
            printf("LED : ON\n");
        }
        else
        {
            printf("LED : OFF\n");
        }

        printf("--------------------------\n");
    }

    return 0;
}
```

Sample Output Screenshot

```
SMART HOME LIGHTING SYSTEM

Enter Motion (1=Detected,0=No Motion): 1
Enter LDR Value (0-1023): 250

Room is Dark
Motion Detected
LED : ON
--------------------------

Enter Motion (1=Detected,0=No Motion): 1
Enter LDR Value (0-1023): 800

LED : OFF
--------------------------
```